



E-R Modelling



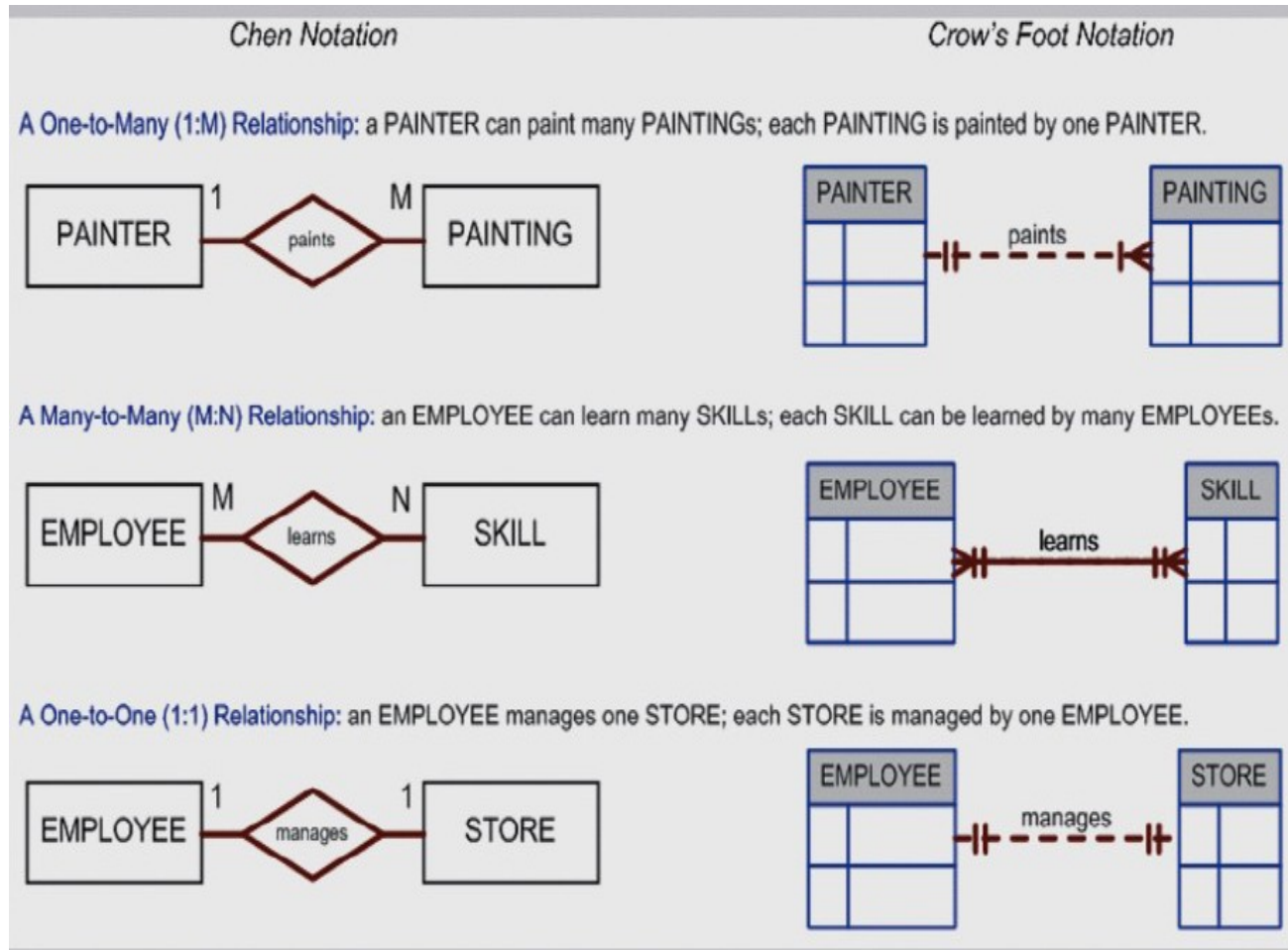
Building Blocks of ERD

Type	English Grammar Equivalent	Example
Entity	Proper Noun	Student, Employee, Instructor, Courses, Room
Relationship	Verb	has, teaches, belongs, handles
Attribute	Adjective	Height, Age, Gender, Nationality, First name

ERD –Two Popular Notation

► Chen Notation

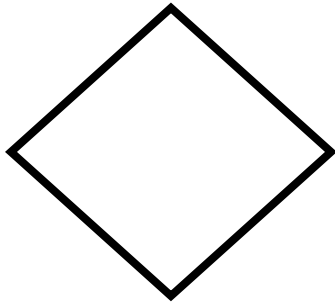
Crow's Foot



Chen Notation - Symbol



Rectangle represents an
Entity



Diamond represents a
Relationship

1



M

Lines with labels represents
Cardinality

Entity (Chen Notation)

- ▶ is a real-world object distinguishable or unique from other objects.
- ▶ An entity can be a **concrete** or **physical** object like *employee, student, faculty, customer* etc. Or it could also be **conceptual** or **abstract** like *transaction, order, course, subjects* etc.
- ▶ It can be thought of as a noun like *student, employee* etc.
- ▶ It is normally represented by a rectangle shape.

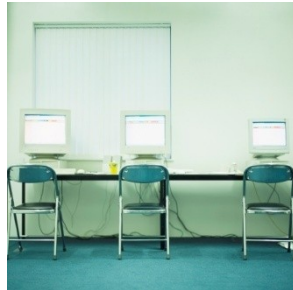


Entity Could be a...



Person

(ex. Teacher, Student, Physician)



Place

(ex. School, Hotel, Store)



Object

(ex. Mouse, Books, Bulding)



Event

(ex. Enroll, Withdraw, Order)



Idea or
Concept

(ex. Courses, Account, Delivery)

Entity - Example

Customer

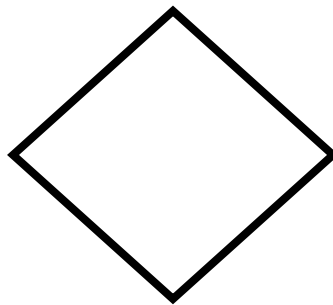
Sales Rep

Order

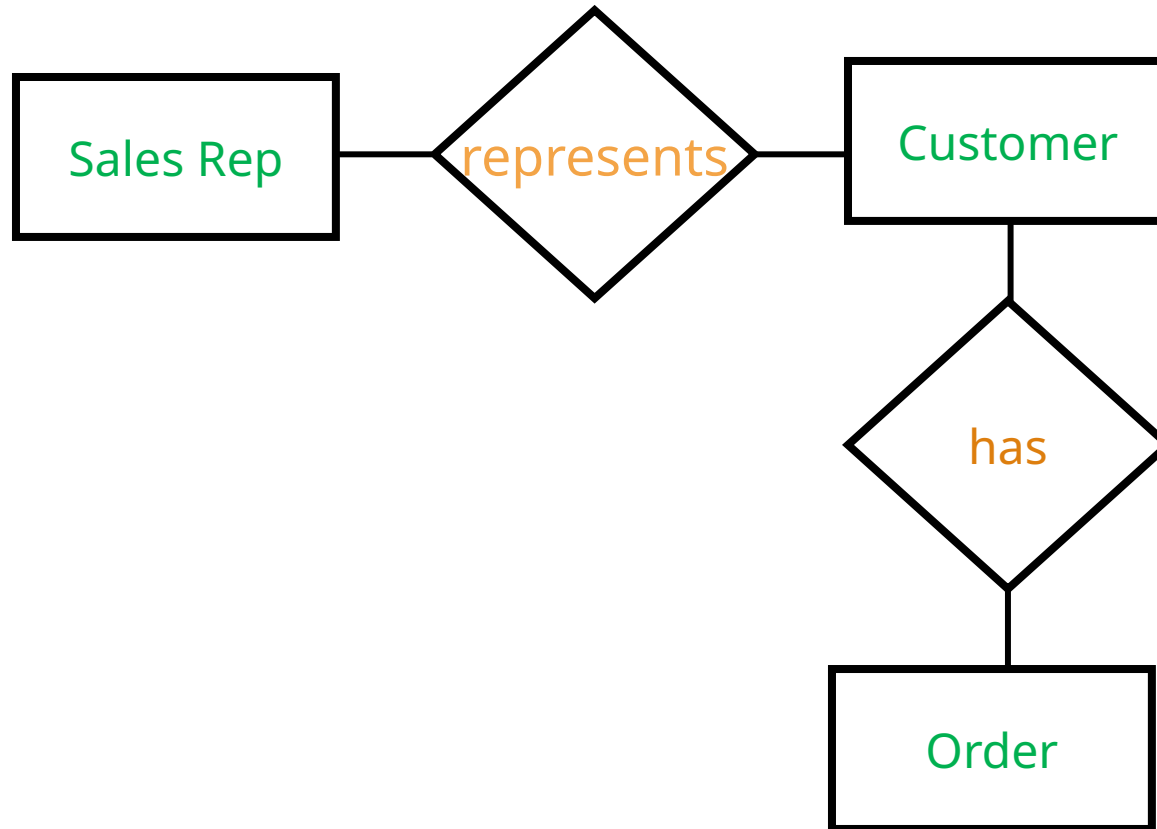
Parts

Relationship

- ▶ is a way of relating one entity to another. Entities can therefore participate in a relationship.
- ▶ it is commonly thought as a *verb* connecting the *entities* or *nouns*.
- ▶ It is normally represented by a diamond shape.



Relationship - Example



*Could be read as : A Sales Rep
Represents a Customer. And*

*a
Customer has an Order.*

Cardinality

- ▶ **Cardinality:** number of items that must be included in a relationship
 - ▶ An entity in a relationship with minimum cardinality of zero plays an **optional role** in the relationship
 - ▶ An entity with a minimum cardinality of one plays a **mandatory role** in the relationship

Cardinality - Symbols

1 _____ 1

One-to-one
Relationship

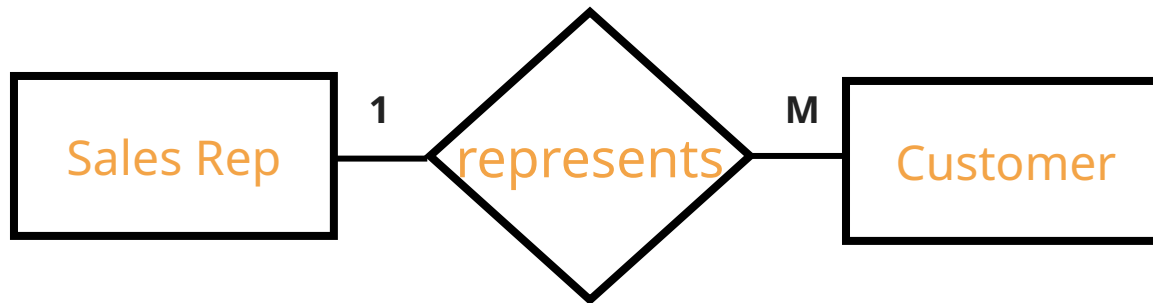
1 _____ M

One-to-many
Relationship

M _____ N

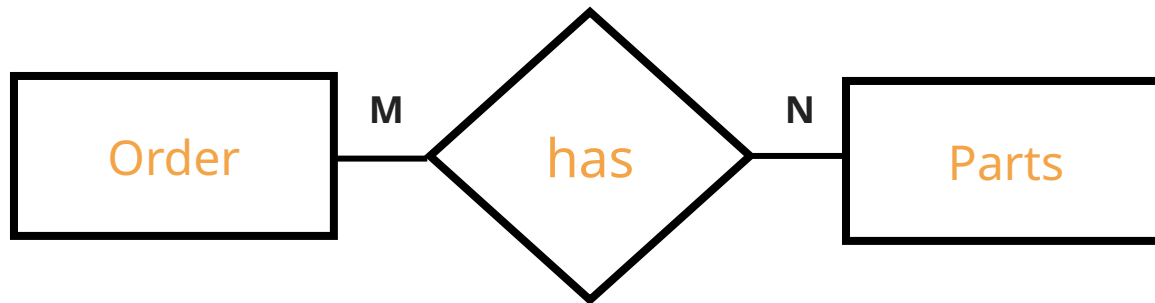
Many-to-many
Relationship

Cardinality Symbols - Example



*Could be read as : A Sales Rep could
represent
1 or Many Customers.*

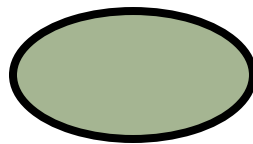
Cardinality Symbols – Example (Cont'd)



Could be read as : An Order could have many Parts (e.g. Products Ordered) and a Part could have many Orders.

Attribute

- ▶ Refers to the characteristic or basic fact or field of an Entity or Relationship.
- ▶ For example a Student entity could have the following **attributes** *ID Number, Last Name, First Name, Address, Birth Date* etc.
- ▶ A relationship could also have an attribute for example an Entity name **Student** enrolls (relationship) to a **Course/Program**. Now, when you enroll you enroll on a certain date so you will have an attribute of *Enrollment Date* under **Enroll** relationship.
- ▶ It is normally represented by an **oval**.



▶ Attribute types:

- ▶ *Simple and composite* (e.g., name (first,middle,last)) attributes
- ▶ *Single-valued and multi-valued* attributes (e.g., multi-valued attribute: *phone-numbers, dependents, etc*)
- ▶ *Derived* attributes
 - ▶ Can be computed from other attributes (e.g., *age*, given date of birth)



Composite Attributes

Composite
Attributes

name

first-name middle-initial last-name

Component
Attributes

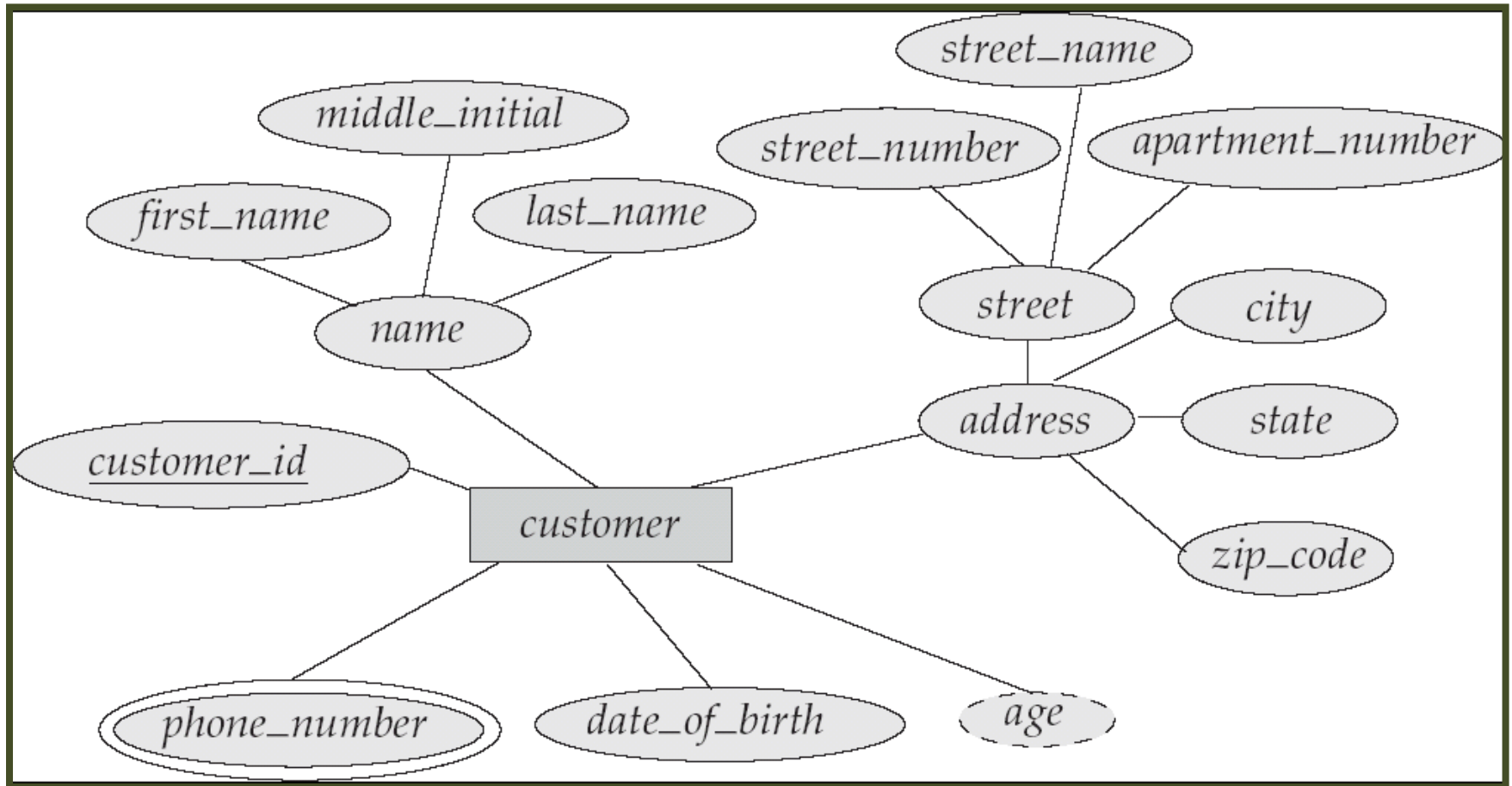
address

street city state postal-code

street-number street-name apartment-number



E-R Diagram With Composite, Multivalued, and Derived Attributes



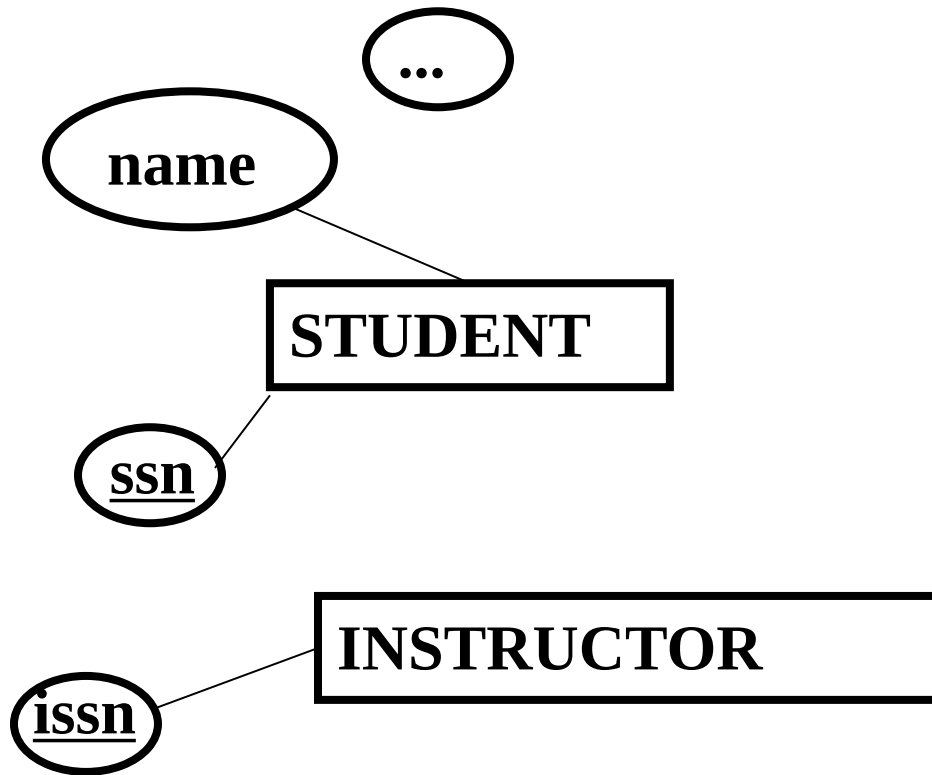
More examples ...

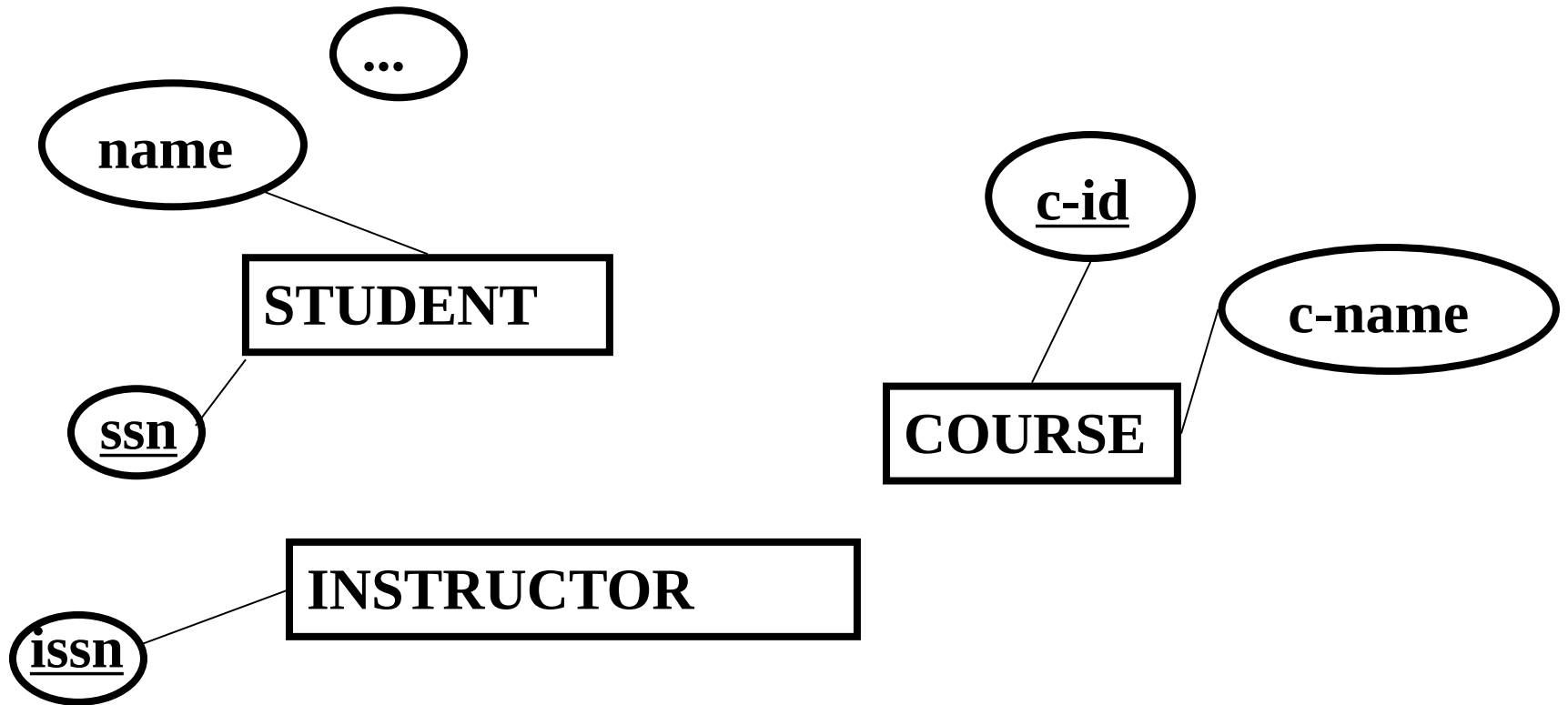
Students, taking courses, offered by instructors;
one instructor per course

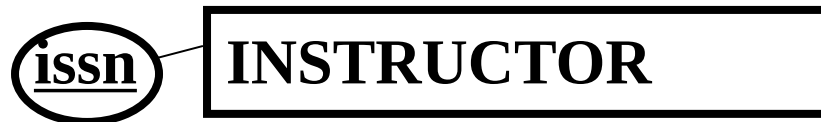
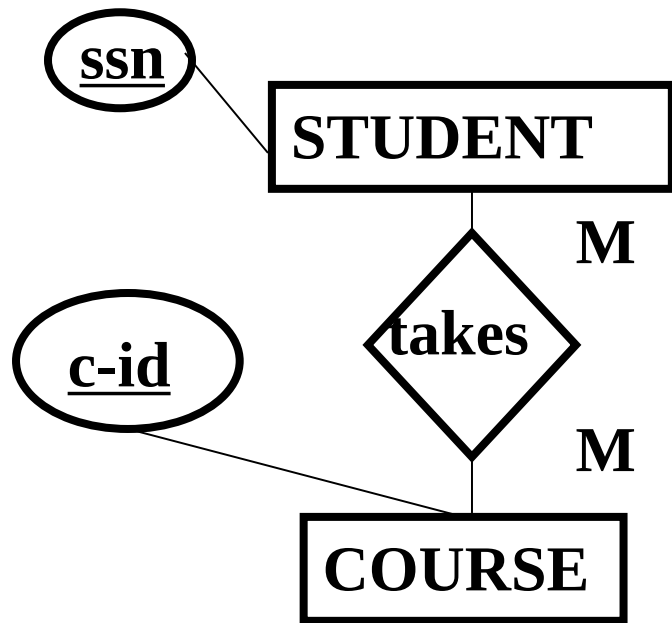
nouns -> entity sets

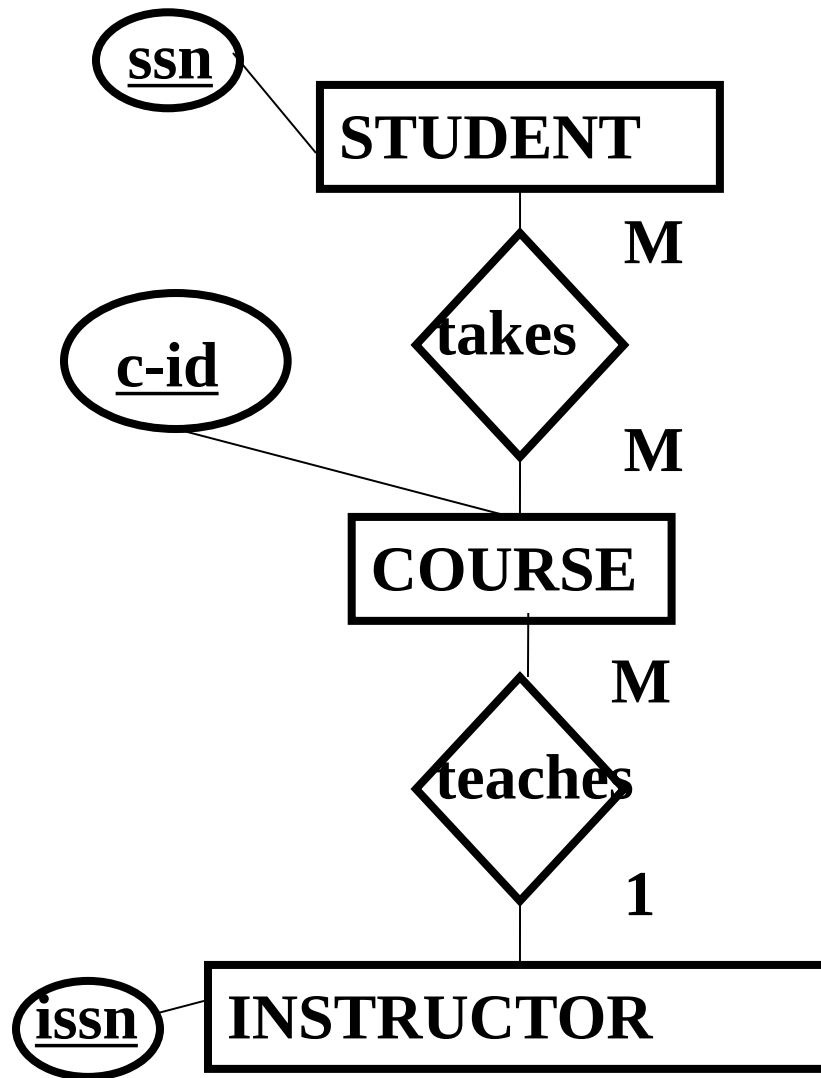
verbs -> relationships









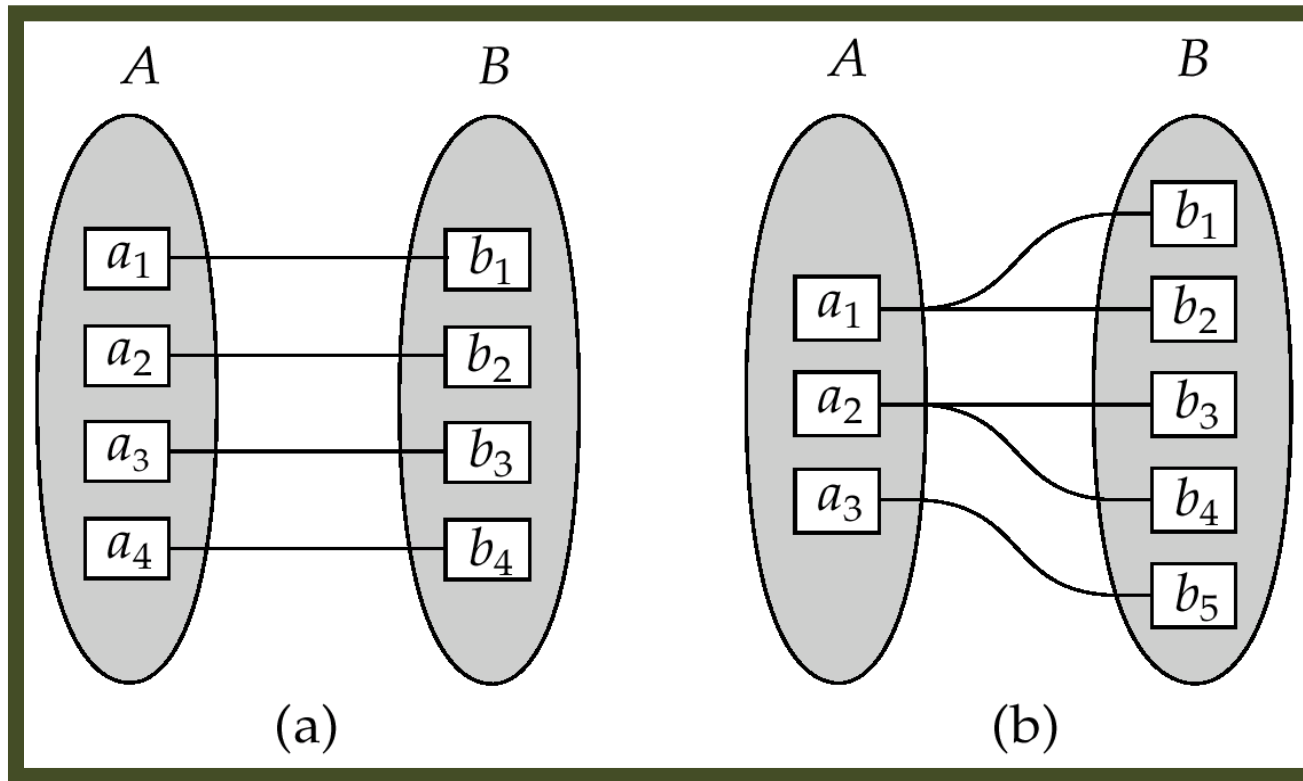


Mapping Constraints: Cardinalities

- ▶ 1 to 1 (example?)
- ▶ 1 to M
- ▶ M to M



Mapping Cardinalities

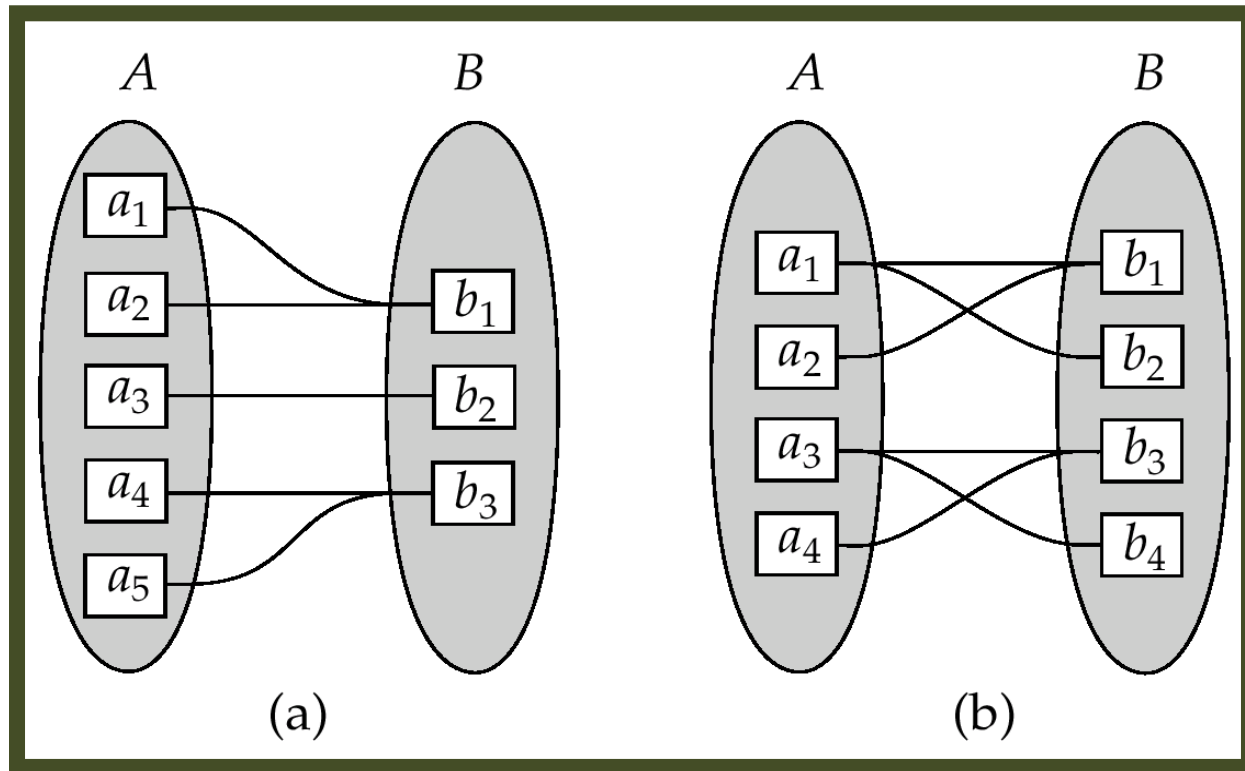


One to one

One to many

Note: Some elements in A and B may not be mapped to any elements in the other set

Mapping Cardinalities

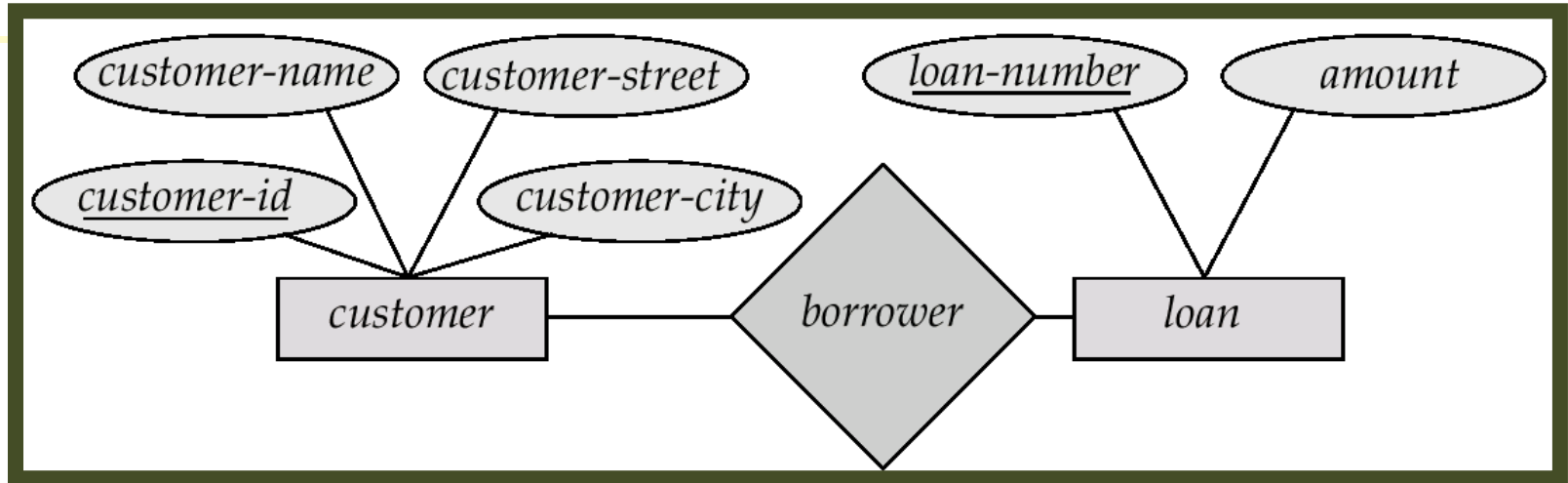


Many to one

Many to many

Note: Some elements in A and B may not be mapped to any elements in the other set

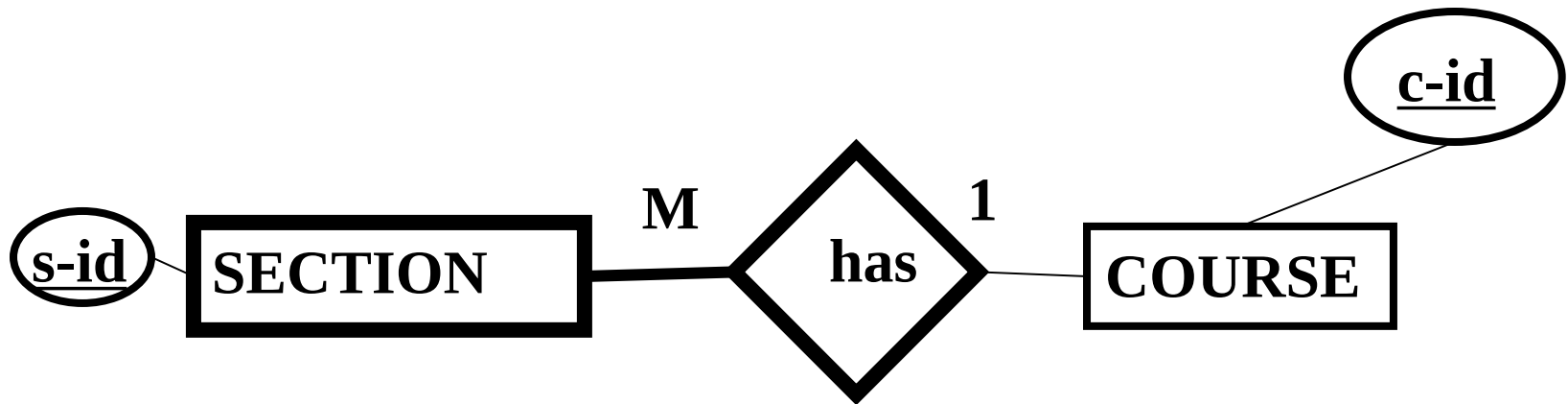
E-R Diagrams



- **Rectangles** represent entity sets
- **Diamonds** represent relationship sets
- **Lines** link attributes to entity sets and entity sets to relationship sets
- **Ellipses** represent attributes
 - **Double ellipses** represent multivalued attributes
 - **Dashed ellipses** denote derived attributes
- **Underline** indicates primary key attributes

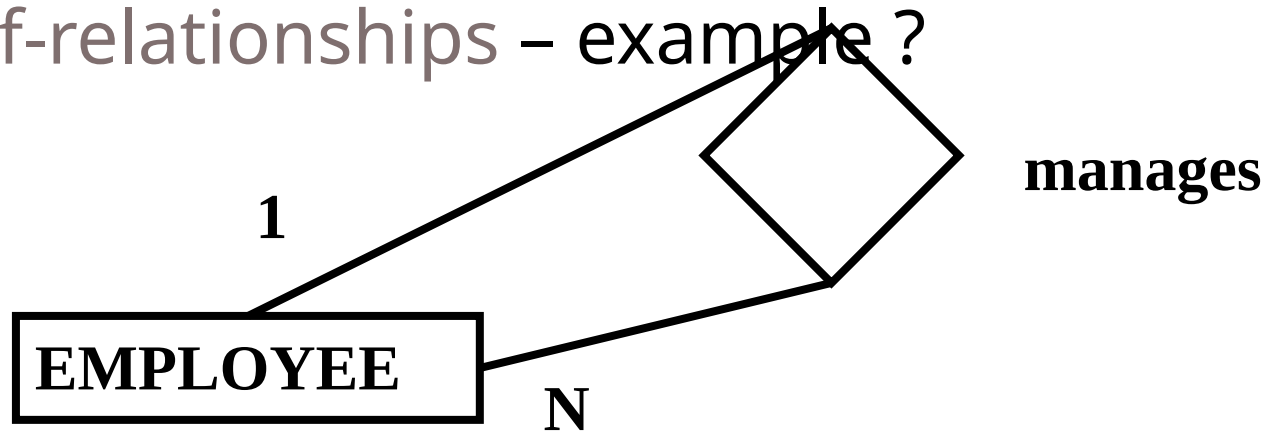
Strong Vs Weak

- 'weak' entities: if they need to borrow a unique id from a 'strong entity - DOUBLE box.
- 'c-id' + 's-id': unique id for SECTION
- discriminator (e.g., 's-id')



More details

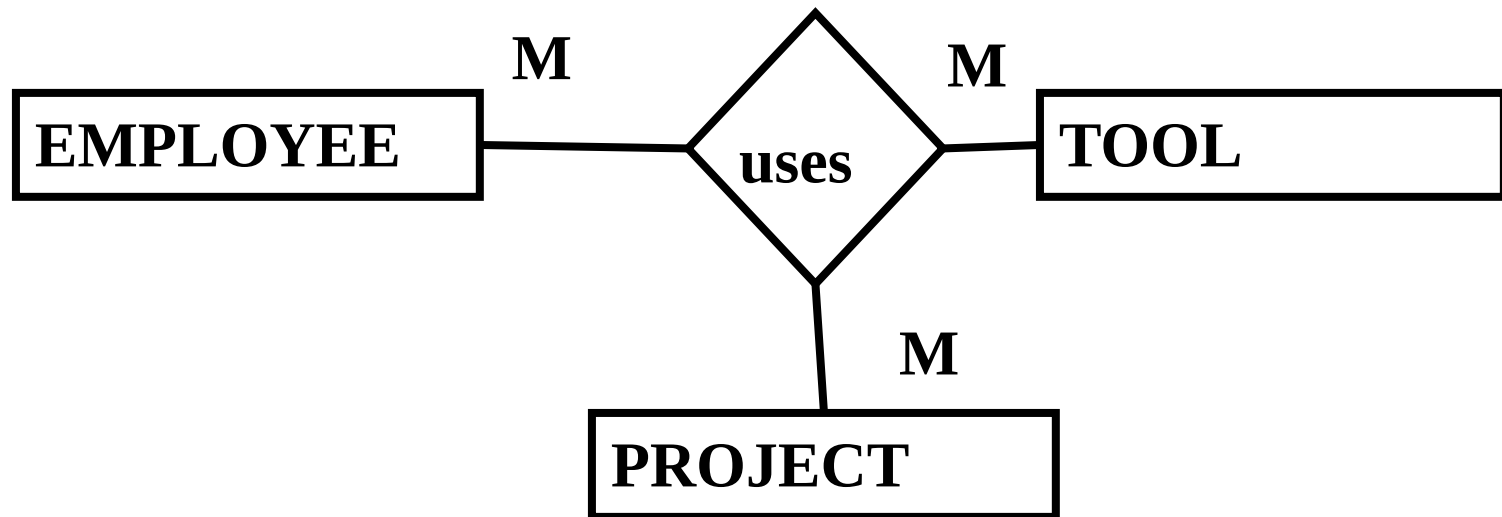
- ▶ Entity sets of a relationship need not be distinct
- ▶ self-relationships – example ?



- ▶ Usually different “roles” are indicated by labeling the lines that connect diamonds to rectangles

More details

- Binary relationships
- 3-way and k-way relationships?



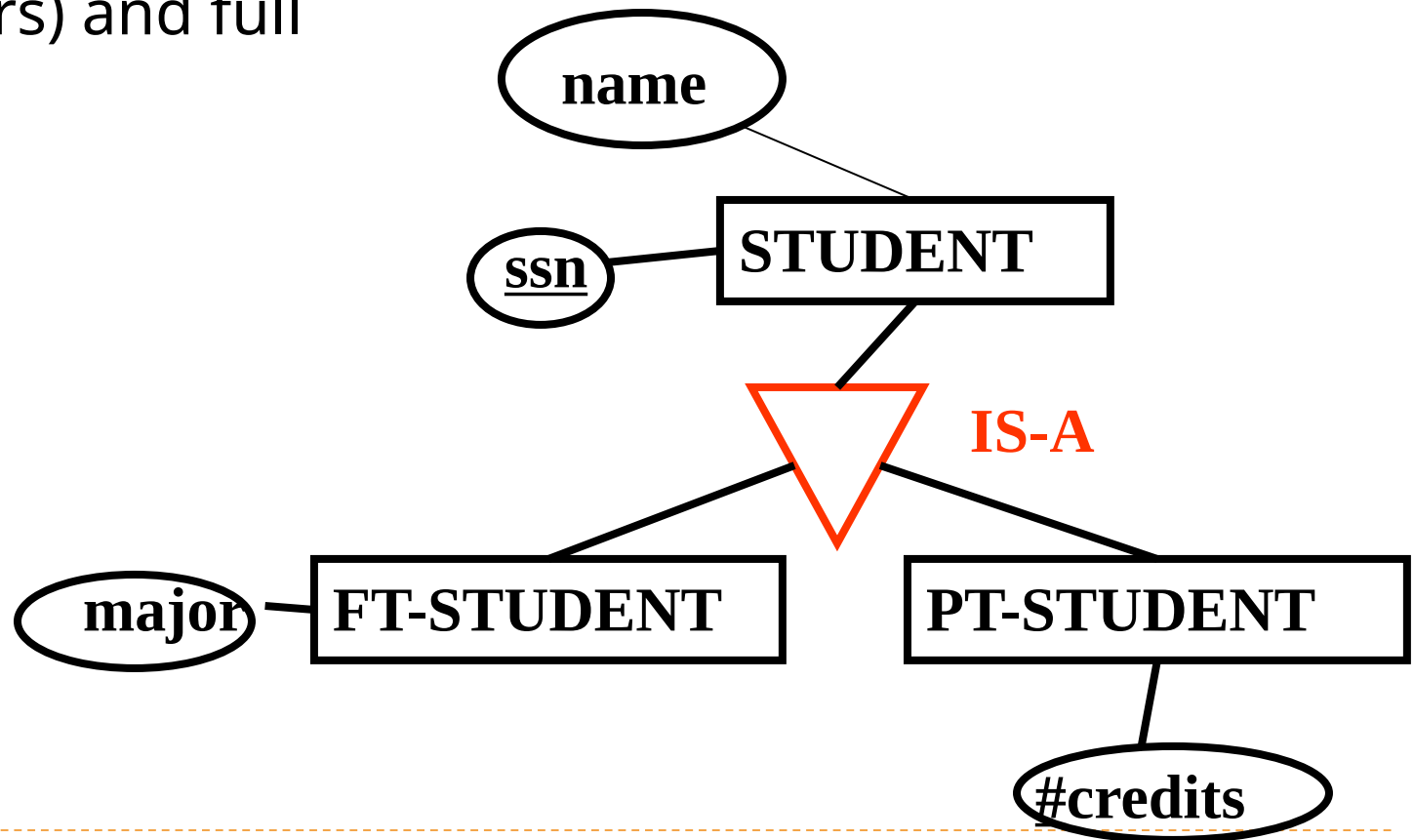
More details - attributes

- ▶ **superkey**: a set of one or more attributes whose values uniquely determine each entity (e.g., (ssn, address))
- ▶ **candidate key**: a minimal super key (e.g., ssn; employee#)
- ▶ **primary key**: a cand. key, chosen by DBA
- ▶ multivalued or set-valued attributes (e.g., 'dependents' for EMPLOYEE)
- ▶ derived attributes (e.g., 15% tip)



Extended ER features: Specialization

e.g., students: part time
(#credit-hours) and full
time (major)

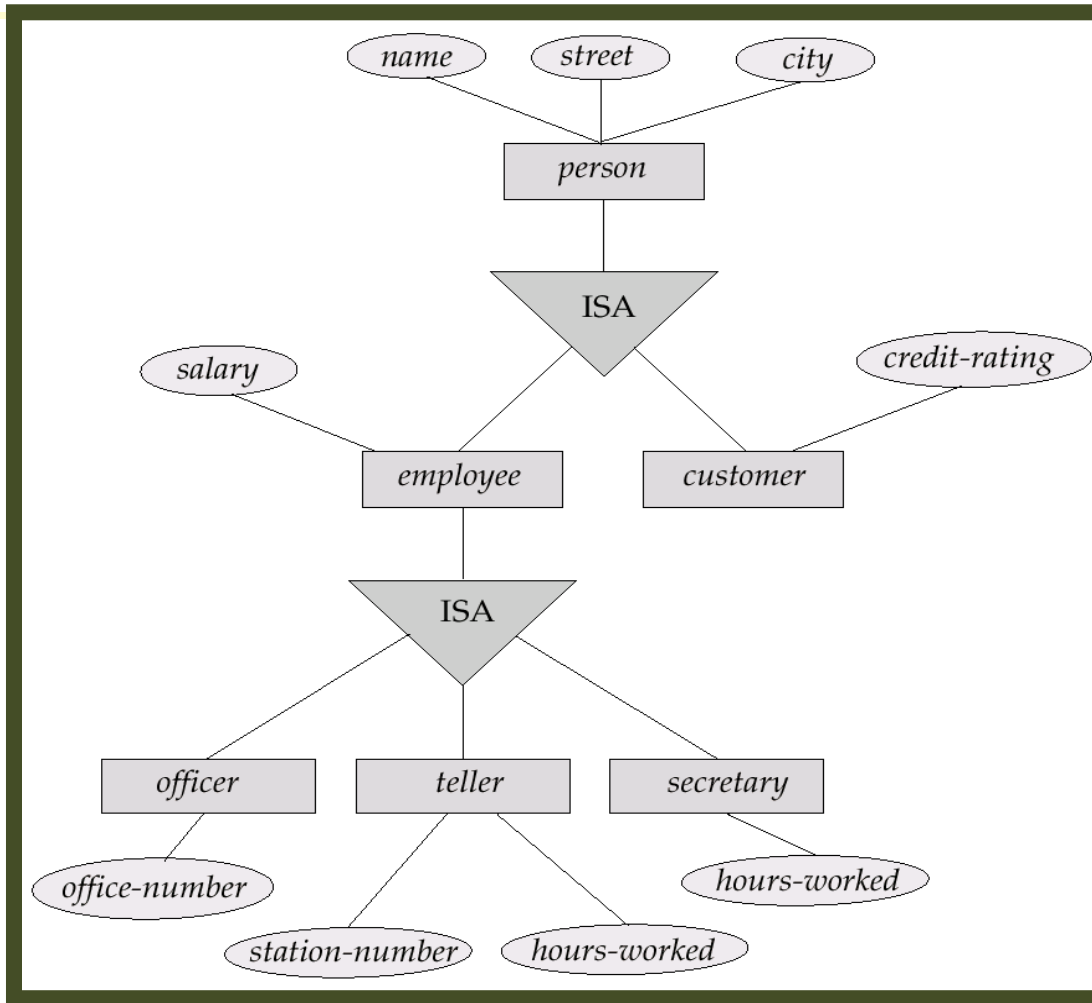


Specialization

- Top-down design process; designate subgroupings within an entity set that are distinctive from other entities in the set
- These subgroupings become lower-level entity sets that have attributes or participate in relationships that do not apply to the higher-level entity set
- Depicted by a *triangle* component labeled **ISA** (E.g. *customer "is a" person*)
- **Attribute inheritance** – a lower-level entity set inherits all the attributes and relationship participation of the higher-level entity set to which it is linked



Another specialization example



Generalization

- ▶ ... opposite to specialization
- ▶ A bottom-up design process – combine a number of entity sets that share the same features into a higher-level entity set
- ▶ Specialization and generalization are inversions of each other; they are represented in an E-R diagram in the same way; we use the terms interchangeably
- ▶ could have **many** levels of an IS-A hierarchy
- ▶ attribute inheritance



Specialization and generalization (Cont.)

- ▶ Can have multiple specializations of an entity set based on different features
- ▶ E.g. *permanent-employee* vs. *temporary-employee*, in addition to *officer* vs. *secretary* vs. *teller*
 - ▶ Each particular employee would be
 - ▶ a member of one of *permanent-employee* or *temporary-employee*,
 - ▶ and also a member of one of *officer*, *secretary*, or *teller*
- ▶ The ISA relationship also referred to as **superclass - subclass** relationship



Design Constraints on Specialization/Generalization

- ▶ The constraint on which entities can be members of a given lower-level entity set:
 - ▶ Condition-defined
 - ▶ E.g. all customers over 65 years are members of *senior-citizen* entity set; *senior-citizen* ISA *person*
 - ▶ User-defined
- ▶ The constraint on whether or not entities may belong to more than one lower-level entity sets within a single generalization
 - ▶ Disjoint
 - ▶ an entity can belong to only one lower-level entity set
 - ▶ Noted in E-R diagram by writing *disjoint* next to the ISA triangle
 - ▶ Overlapping
 - ▶ an entity can belong to more than one lower-level entity set

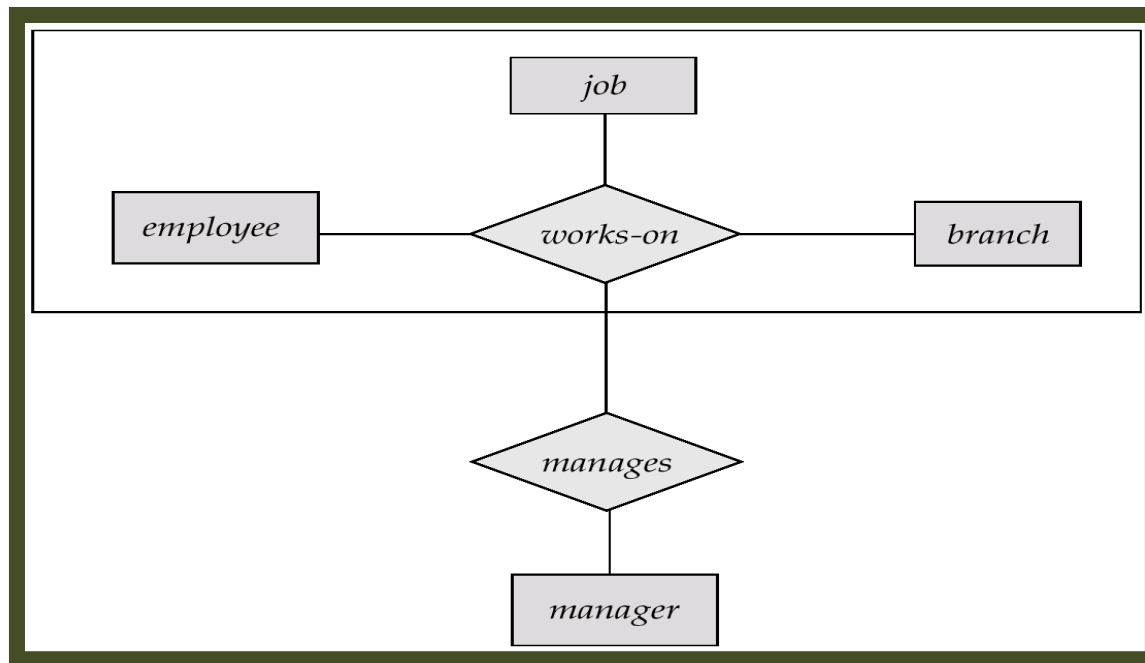
Design Constraints on a Specialization/Generalization

- ▶ **Completeness constraint** -- specifies whether or not an entity in the higher-level entity set must belong to at least one of the lower-level entity sets within a generalization
- ▶ **total** : an entity must belong to one of the lower-level entity sets
- ▶ **partial**: an entity need not belong to one of the lower-level entity sets

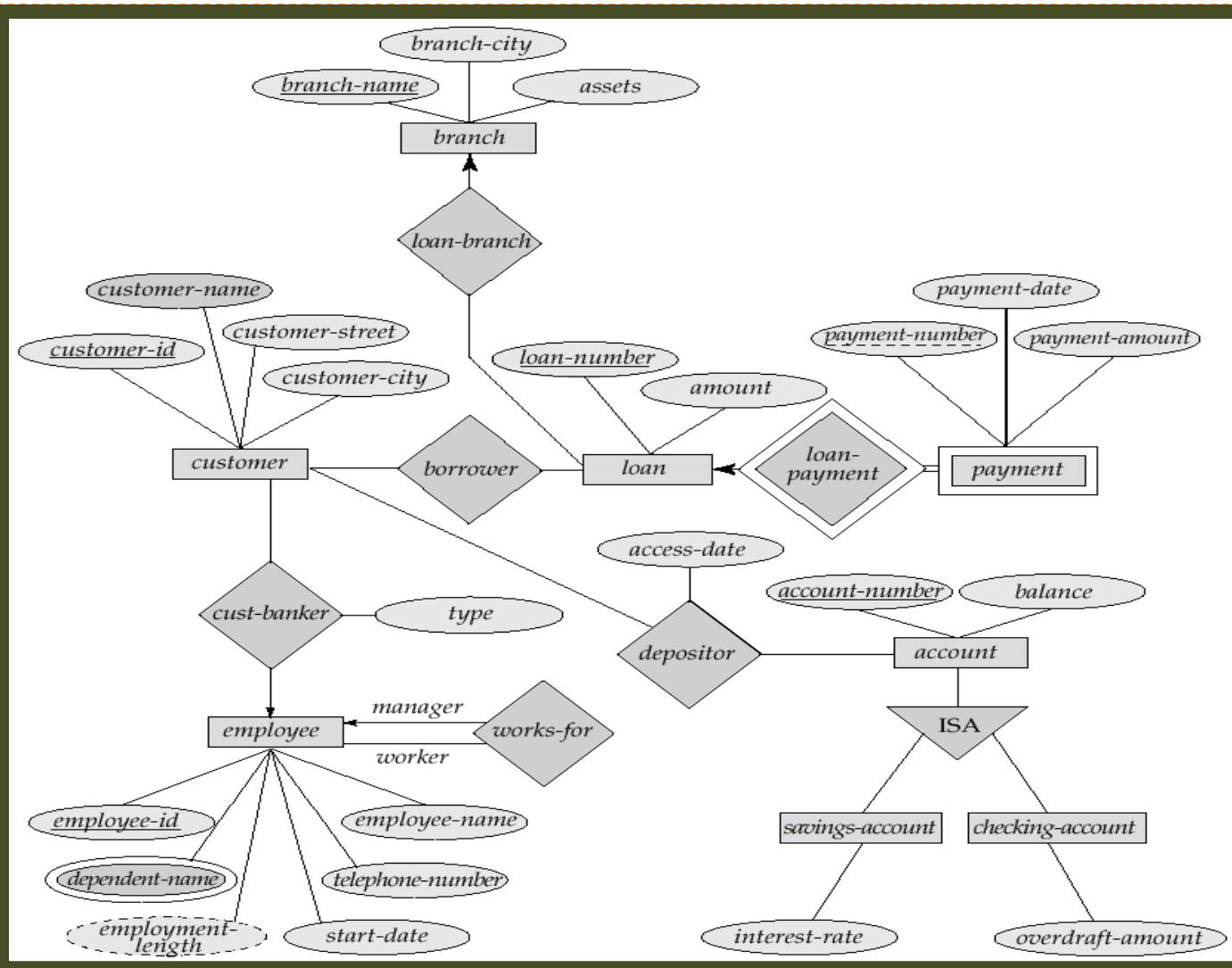


Aggregation

treats a relationship as an 'abstract' entity
allows relationships between relationships
rarely used



E-R Diagram for a Banking Enterprise





Reduction of E-R Diagrams to Tables



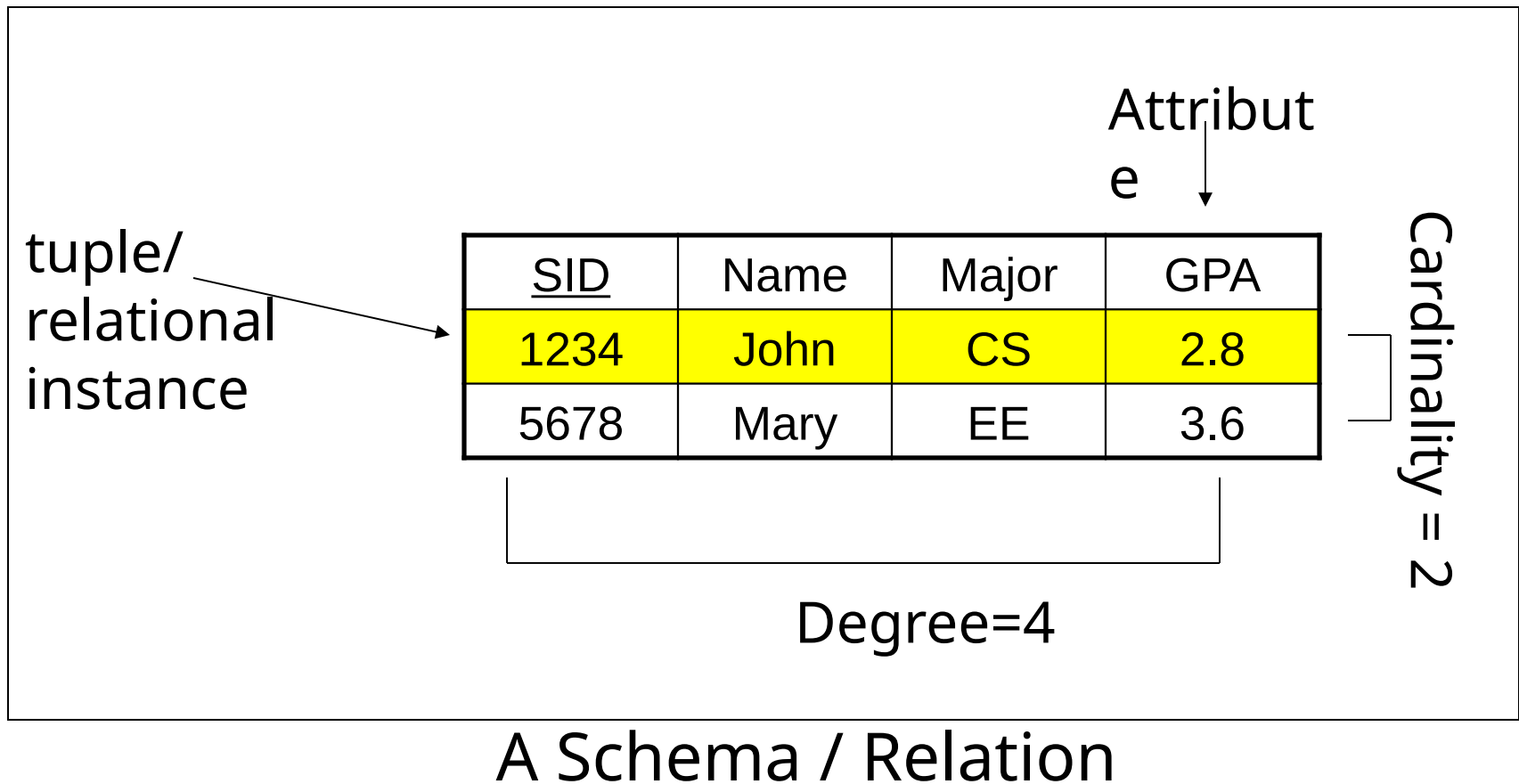
Relational Model

Relational Model is made up of tables

- ▶ A row of table = a relational instance/tuple
- ▶ A column of table = an attribute
- ▶ A table = a schema/relation
- ▶ Cardinality = number of rows
- ▶ Degree = number of columns



Example



From ER Model to Relational Model

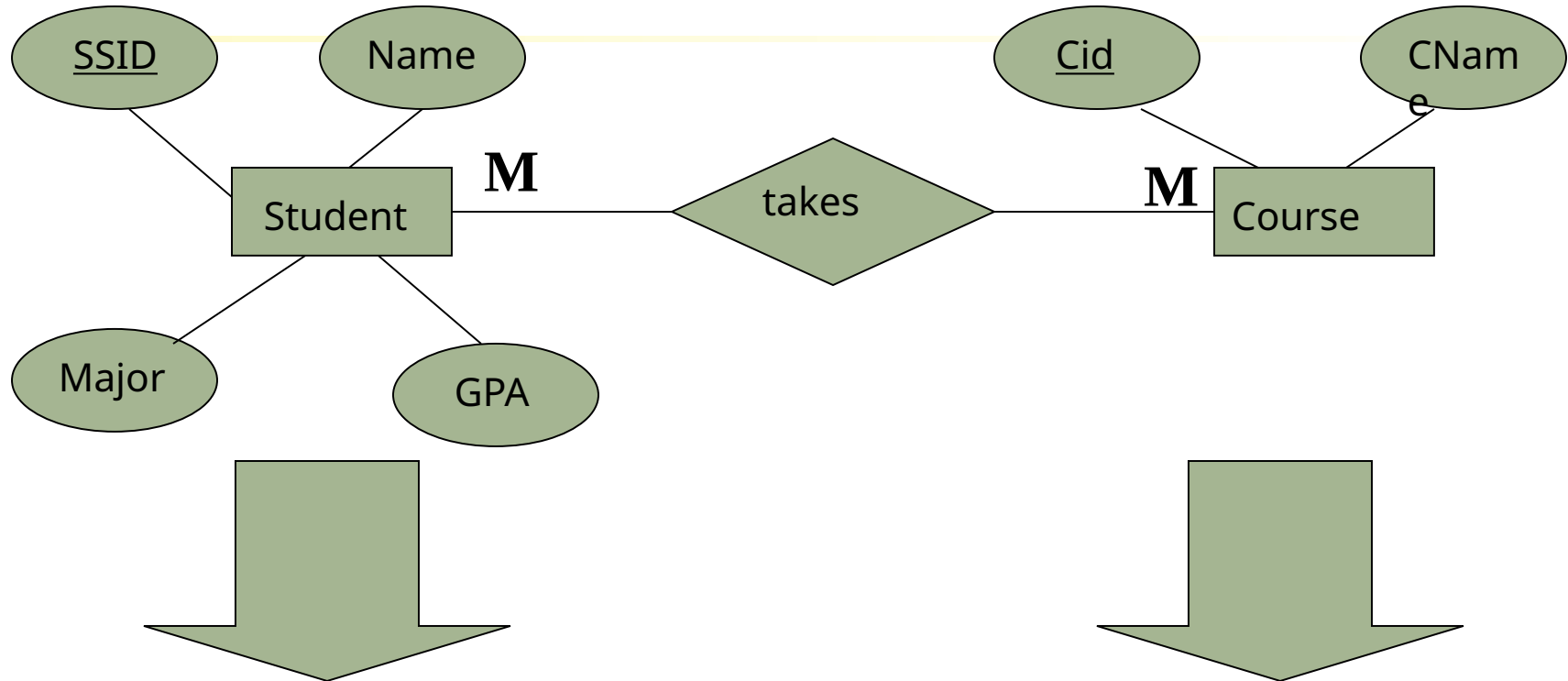
So... how do we convert an ER diagram into a table?? Simple!!

Basic Ideas:

- Build a table for each entity set
- Make a column in the table for each attribute in the entity set
- Primary Key
- Build a table for each relationship set if necessary



Example – Strong Entity Set



<u>SID</u>	Name	Major	GPA
1234	John	CS	2.8
5678	Mary	EE	3.6

<u>Cid</u>	CName
C1	C
C2	Java

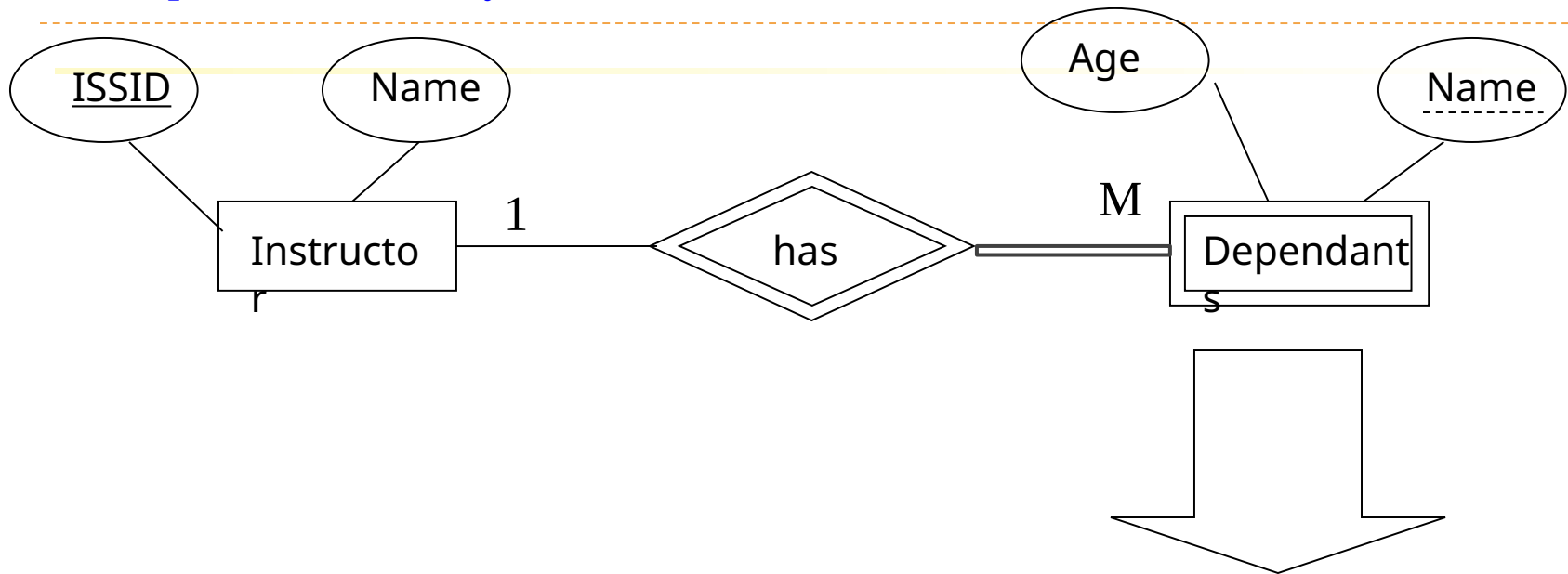


Representation of Weak Entity Set

- ▶ Weak Entity Set Cannot exist alone
- ▶ To build a table/schema for weak entity set
 - ▶ Construct a table with one column for each attribute in the weak entity set
 - ▶ Remember to include discriminator
 - ▶ Augment one extra column on the right side of the table, put in there the primary key of the Strong Entity Set (the entity set that the weak entity set is depending on)
 - ▶ Primary Key of the weak entity set = Discriminator + foreign key



Example – Weak Entity Set



Age	Name	<u>Parent_ISSID</u>
10	Bart	1234
8	Lisa	5678

* Primary key of *Children* is *Parent_ISSID* + *Name*

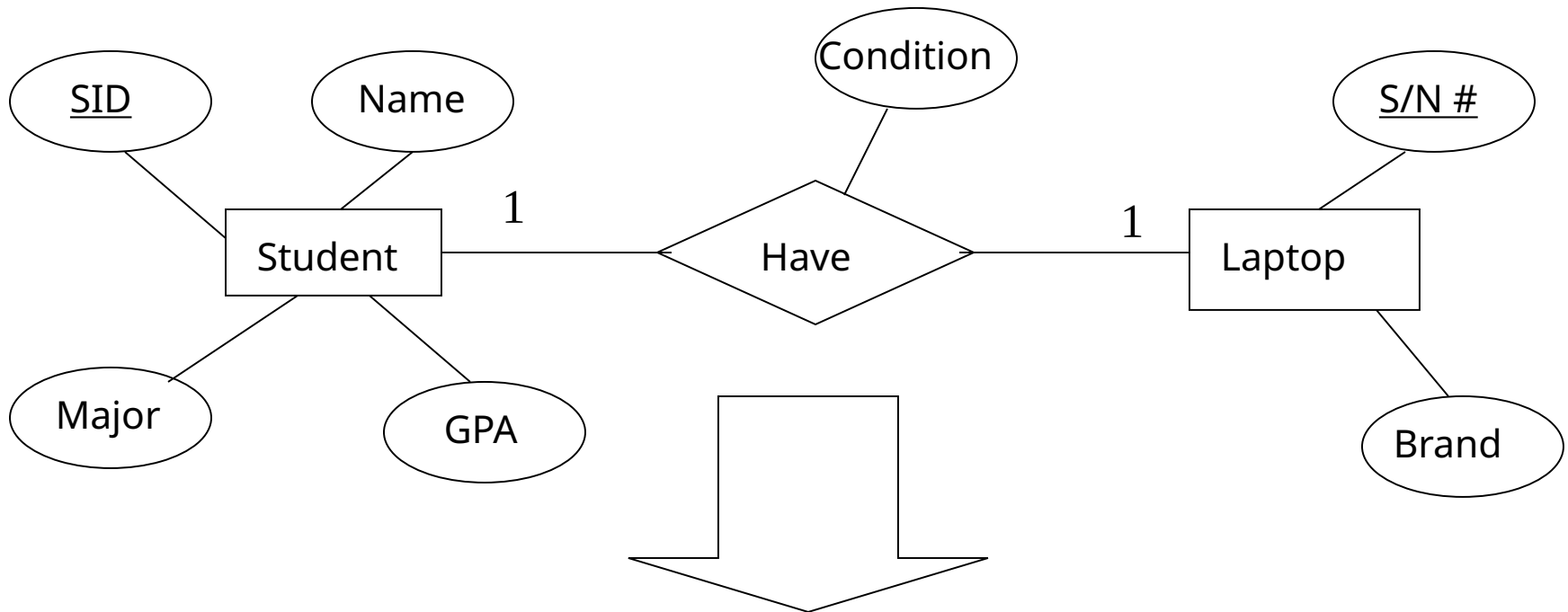
Representation of Relationship Set

- ▶ For one-to-one relationship sets, either side can be chosen to act as the “many” side
 - ▶ That is, extra attribute can be added to either of the tables corresponding to the two entity sets



Example – One-to-One Relationship Set

1:1 Relationship



<u>SID</u>	Name	Major	GPA	LP_S/N	Hav_Cond
9999	Bart	Economy	-4.0	123-456	Own
8888	Lisa	Physics	4.0	567-890	Loan

* Primary key can be either *SID* or *LP_S/N*

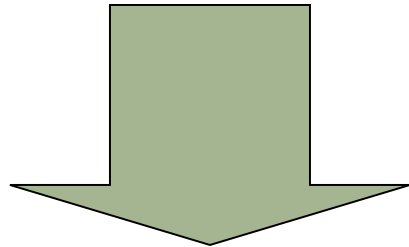
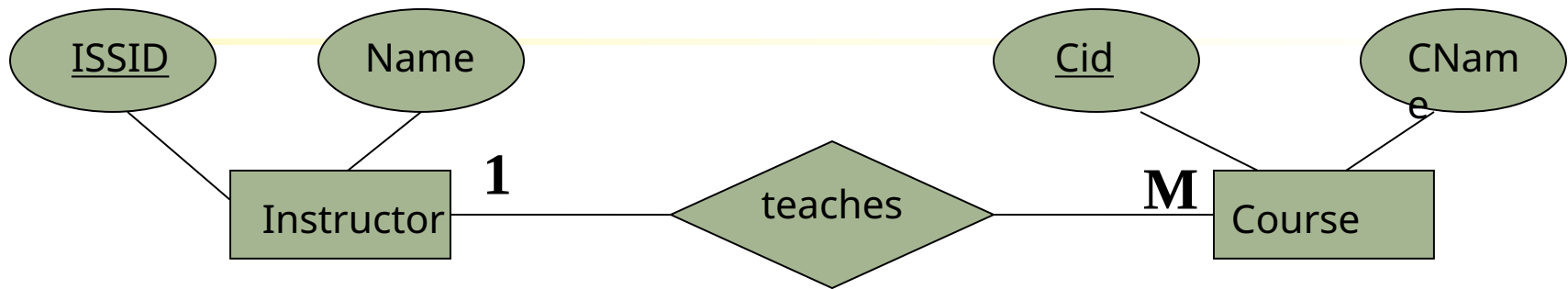
Representing Relationship Set

- Many-to-one and one-to-many relationship sets can be represented by adding an extra attribute to the “many” side, containing the primary key of the “one” side

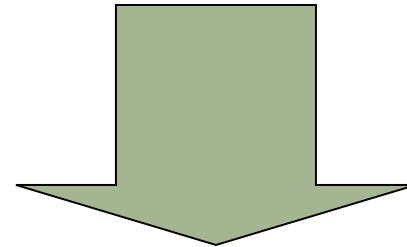




Example – Many-to-One Relationship Set



<u>ISSID</u>	Name
12	Joan
56	Martha



<u>Cid</u>	CName	ISSID
C1	C	12
C2	Java	56

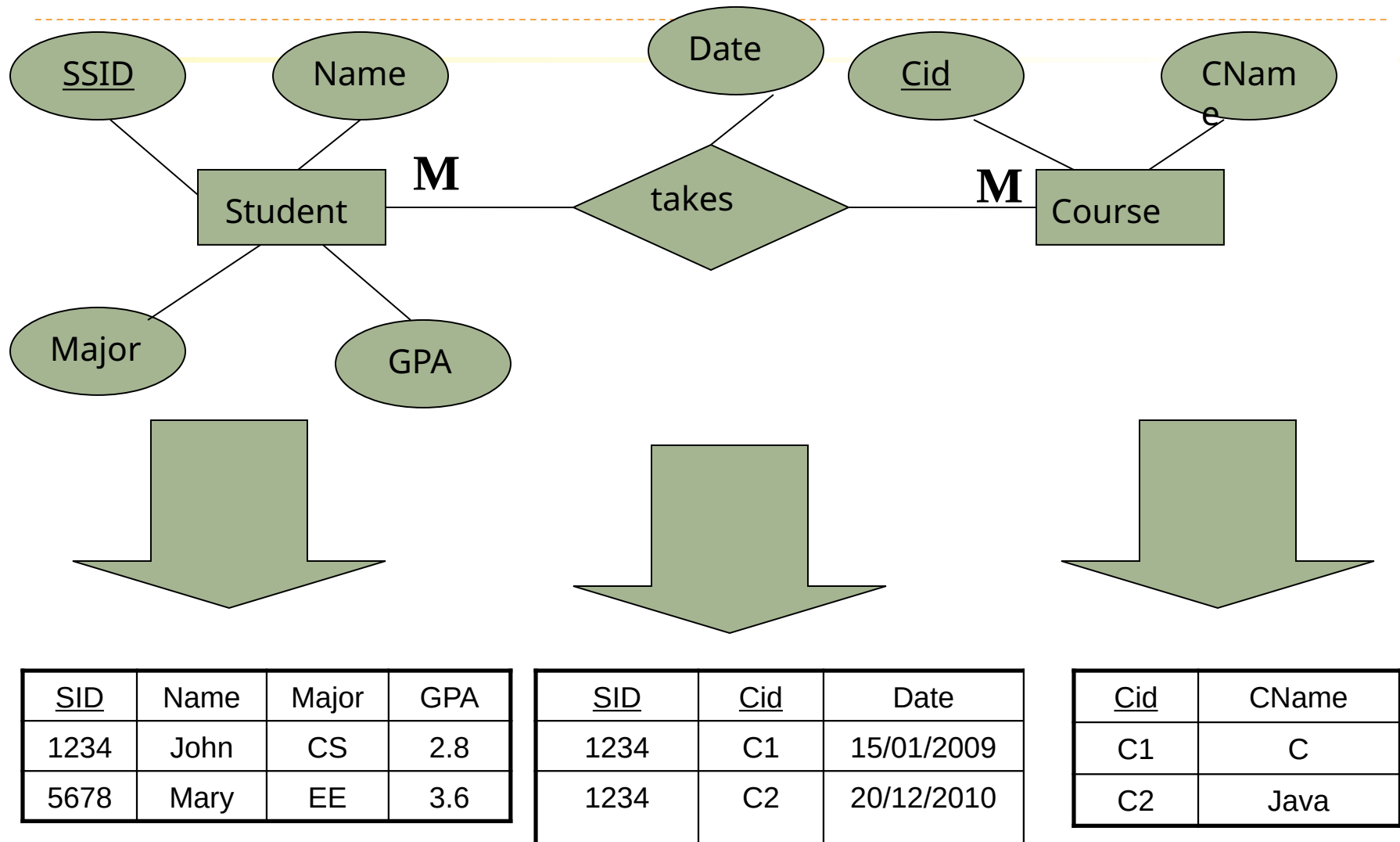


Representing Relationship Set

- ▶ For many-to-many relationship
 - ▶ Primary key of this new schema is the union of the foreign keys of both entity sets.

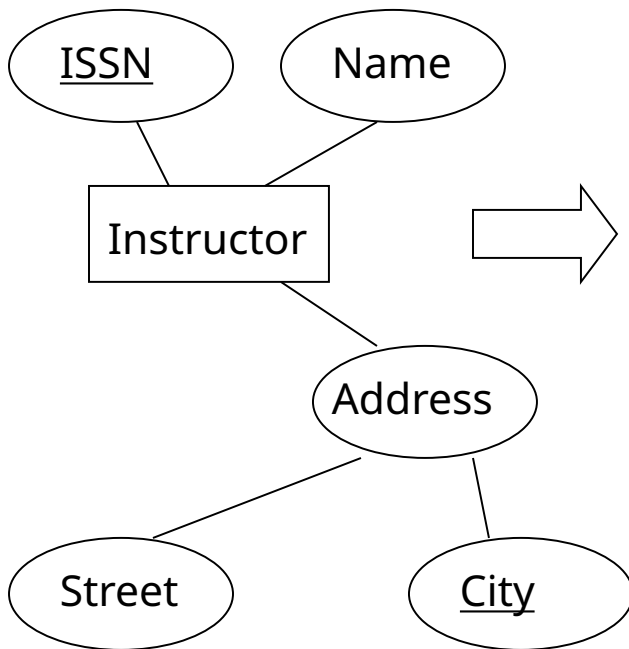


Example – Many-to –Many Relation Ship set



Representing Composite Attribute

- ▶ One column for each component attribute
- ▶ NO column for the composite attribute itself



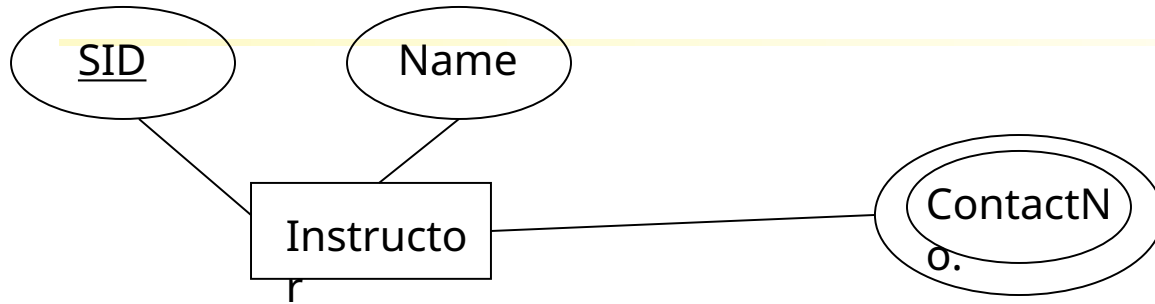
<u>ISSN</u>	Name	Street	City
99	Dr. Smith	50 1 st St.	Fake City
88	Dr. Lee	1 B St.	San Jose

Representing Multivalued Attribute

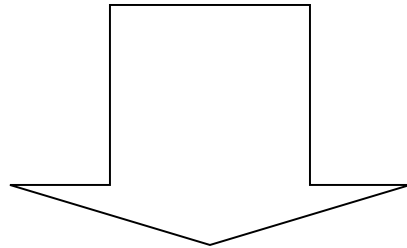
- ▶ For each multivalued attribute in an entity set/relationship set
 - ▶ Build a new relation schema with two columns
 - ▶ One column for the primary keys of the entity set/relationship set that has the multivalued attribute
 - ▶ Another column for the multivalued attributes. Each cell of this column holds only one value. So each value is represented as a unique tuple
 - ▶ Primary key for this schema is the union of all attributes



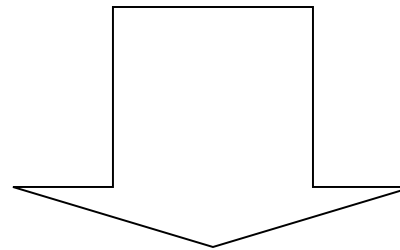
Example – Multivalued attribute



The primary key for this table is Student_SID + Children, the union of all attributes



<u>ISSID</u>	Name	Major	GPA
12	Joan	CS	2.8
56	Martha	EE	3.6



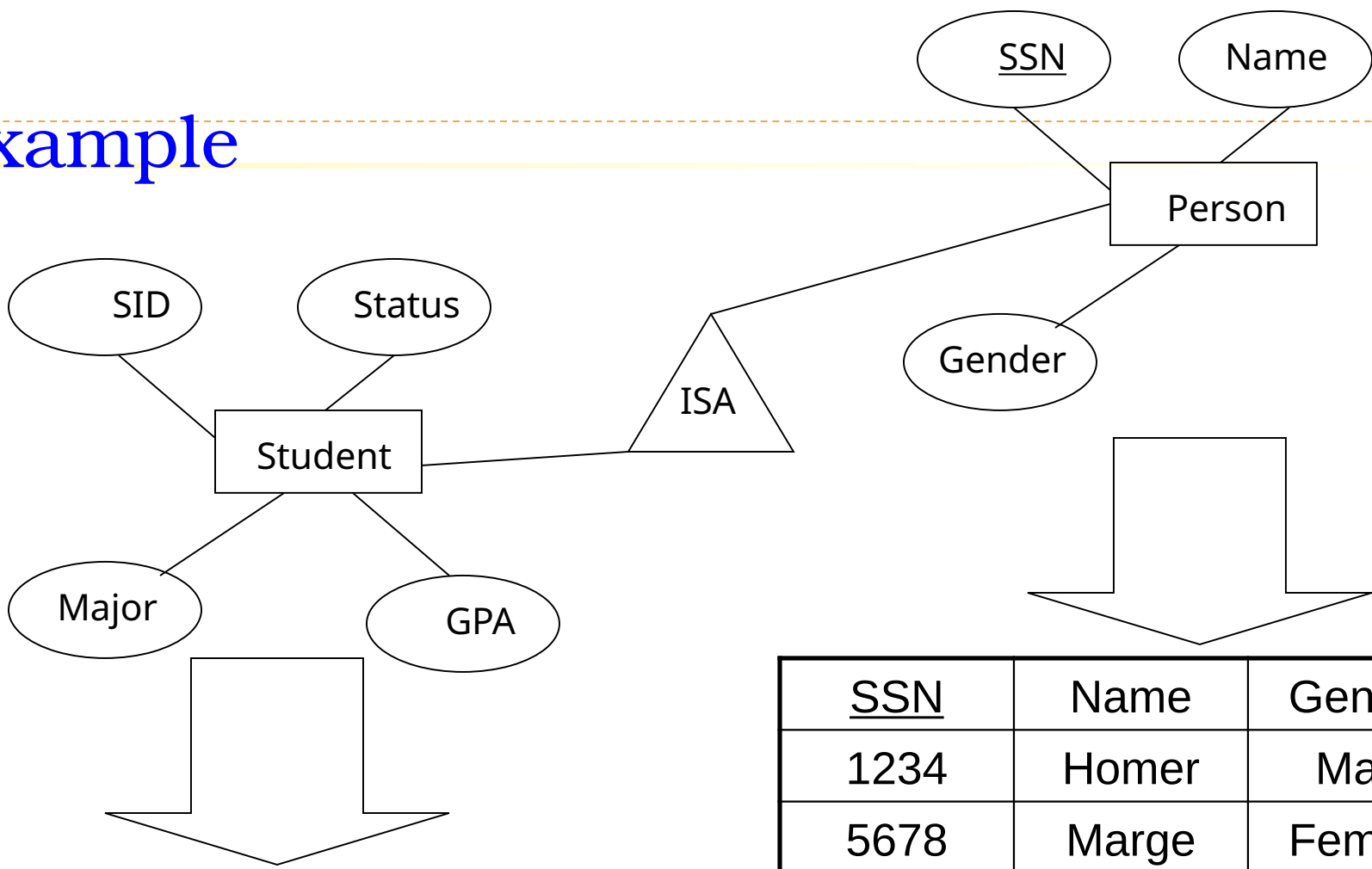
<u>ISSID</u>	ContactNo
12	222222
12	345677
56	777799
56	998765
56	333455

Representing Class Hierarchy

- ▶ Two general approaches depending on disjointness and completeness
 - ▶ For non-disjoint and/or non-complete class hierarchy:
 - ▶ create a table for each super class entity set according to normal entity set translation method.
 - ▶ Create a table for each subclass entity set with a column for each of the attributes of that entity set plus one for each attributes of the primary key of the super class entity set
 - ▶ This primary key from super class entity set is also used as the primary key for this new table



Example



<u>SSN</u>	Name	Gender
1234	Homer	Male
5678	Marge	Female

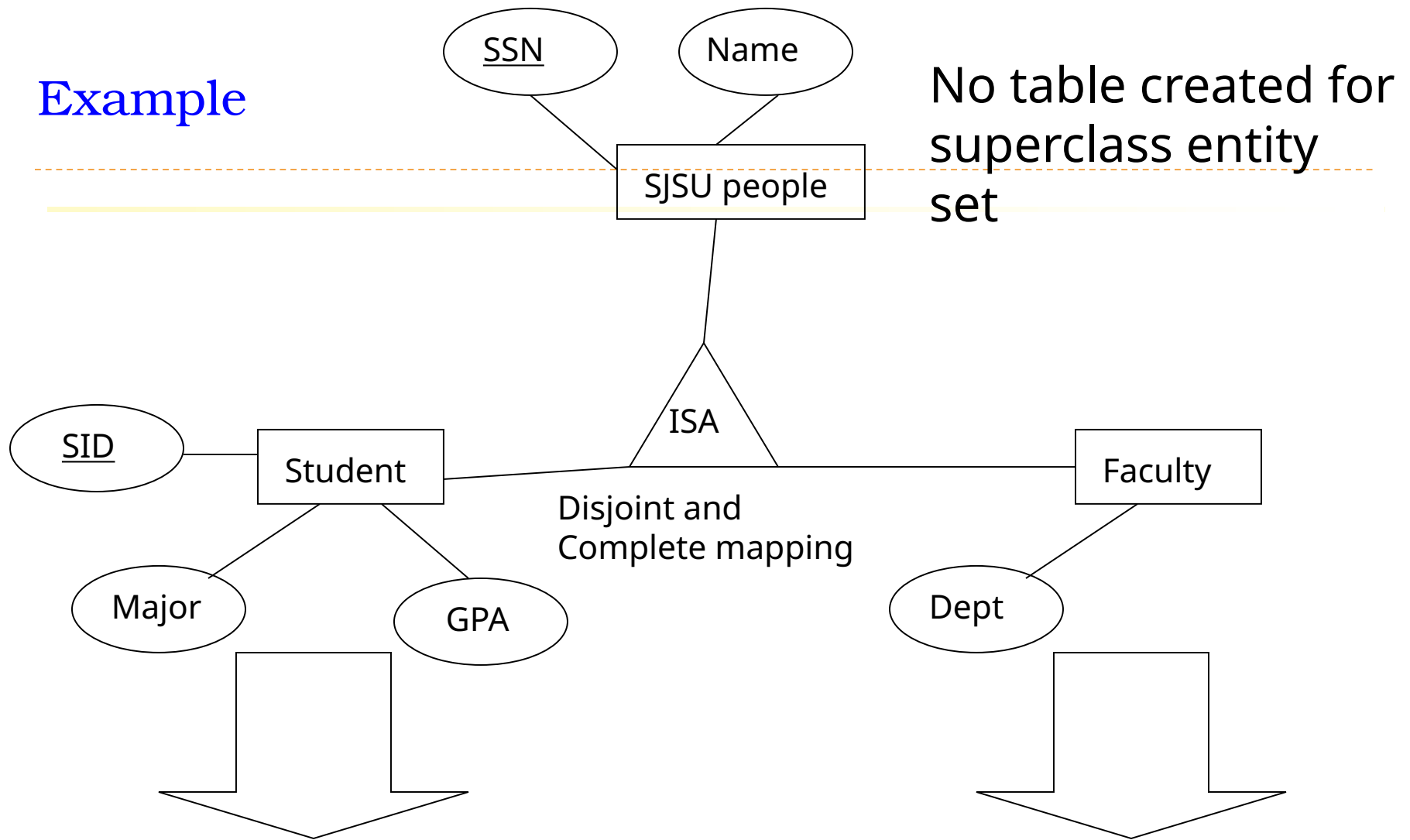
<u>SSN</u>	SID	Status	Major	GPA
1234	9999	Full	CS	2.8
5678	8888	Part	EE	3.6

Representing Class Hierarchy

- ▶ Two general approaches depending on disjointness and completeness
 - ▶ For disjoint **AND** complete mapping class hierarchy:
 - ▶ DO NOT create a table for the super class entity set
 - ▶ Create a table for each subclass entity set include all attributes of that subclass entity set and attributes of the superclass entity set
- ▶ Simple and Intuitive enough, need example?



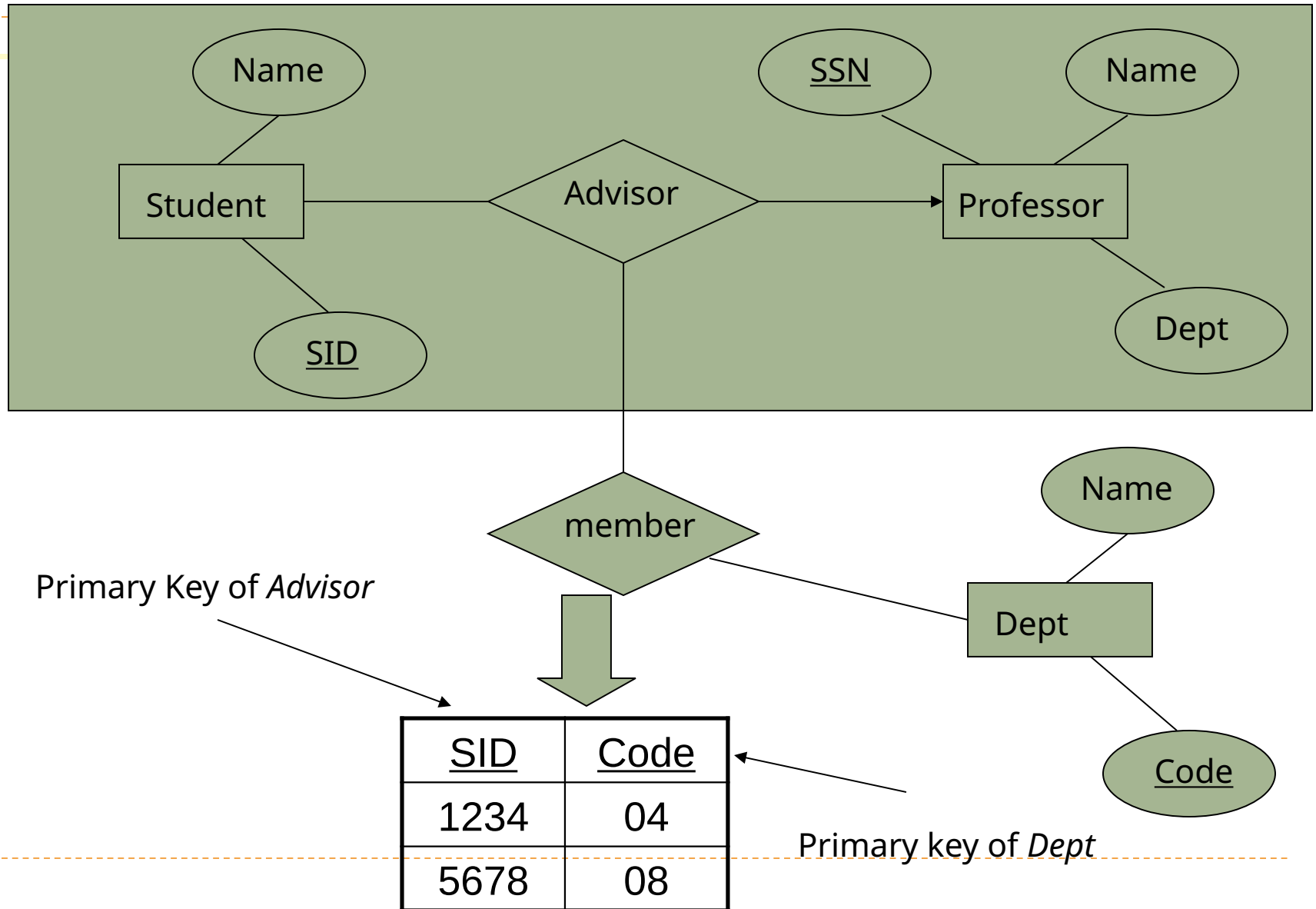
Example



<u>SSN</u>	Name	SID	Major	GPA
1234	John	9999	CS	2.8
5678	Mary	8888	EE	3.6

<u>SSN</u>	Name	Dept
1234	Homer	C.S.
5678	Marge	Math

Representing Aggregation



Representing aggregation

- ▶ make table, with primary keys of all involved entities





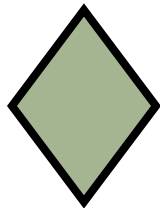
Summary - cont'd



(strong) entity set



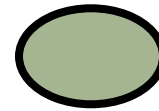
weak entity set



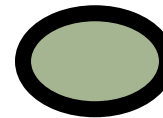
relationship set



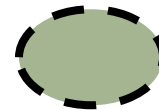
**identifying rel. set
for weak entity**



attribute



**multivalued
attribute**



**derived
attribute**



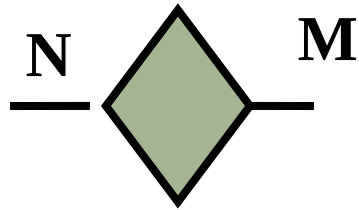
Summary - cont'd



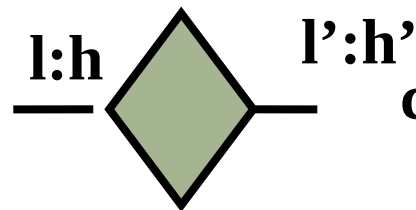
primary key



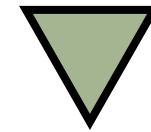
discriminator



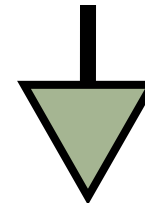
cardinalities



**cardinalities
with limits**



IS-A



**total
generalization**

